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1. No work

$$P = F/A$$

2. $d = 5$

$$r = 2.5$$

$$A = 2.5^2 \cdot \pi$$

$$A = 19.63495$$

$$A = 19.6 \text{ in}^2$$

3. $(39.2) 250016 \text{ lb} = \frac{F16}{2 \cdot 19.6} \left(\frac{39.2}{1} \right)$

$$98171.77 = F$$

$$9810015$$

4. No work

5. $21.0 \text{ gal/min} \left| \frac{1000 \text{ cm}^3}{0.264 \text{ gal}} \right| \frac{1 \text{ inch}^3}{2.54 \text{ cm}^3} = 4854.161462 \text{ in}^3/\text{min}$
 $4850 \text{ in}^3/\text{min}$

6.

$$d = 1.75$$

$$r = 0.875$$

$$A = 0.875^2 \cdot \pi$$

$$A = 2.4052$$

$$A = 2.41 \text{ in}^2$$

7. $\frac{4854.16}{2.41} = \frac{fV \cdot 2.41}{2.41}$

$$2018.125 = fV$$

$$2020 \text{ in/min}$$

8. $MA = 825/61.75/5 = 2.65$

$$MA = 11.25$$

$$P_{PF} = 75/10 = 7.5 \text{ PSI}$$

9. $P_o(15)7.5 = F/15(15)$ $825/112.5 = 7.3$

$$112.5 = F/15$$

$$8 \text{ cylinders}$$